

CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

Linear Transformations

Worked solutions

11 questions · 85 marks

Questions

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1. (i) In each of the following cases, find a 2×2 matrix that represents
- (a) a reflection in the line $y = x$ [1]
 - (b) a rotation of 60° clockwise about $(0, 0)$ [1]
 - (c) a rotation of 60° clockwise about $(0, 0)$ followed by a reflection in the line $y = x$ [2]
- (ii) The triangle T has vertices at the points $(1, 2)$, $(5, 5)$ and $(k, -1)$, where $k > -3$ is a constant.

Triangle T is transformed onto the triangle T' by the matrix

$$\begin{pmatrix} 2 & 5 \\ 1 & -1 \end{pmatrix}$$

Given that the area of triangle T' is 84 square units, find the value of k . [6]

Solution

- (i) For a 2×2 transformation matrix, the first column is the image of $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and the second column is the image of $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

- (a) A reflection in the line $y = x$ swaps the coordinates, so

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Hence the matrix is

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

- (b) A clockwise rotation through angle θ about the origin has matrix

$$\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

With $\theta = 60^\circ$,

$$\cos 60^\circ = \frac{1}{2}, \quad \sin 60^\circ = \frac{\sqrt{3}}{2}$$

So the matrix is

$$\begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

- (c) Let

$$R = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Since the rotation is done first and the reflection second, the required matrix is FR .

$$\begin{aligned} FR &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \\ &= \begin{pmatrix} 0 \cdot \frac{1}{2} + 1 \cdot \left(-\frac{\sqrt{3}}{2}\right) & 0 \cdot \frac{\sqrt{3}}{2} + 1 \cdot \frac{1}{2} \\ 1 \cdot \frac{1}{2} + 0 \cdot \left(-\frac{\sqrt{3}}{2}\right) & 1 \cdot \frac{\sqrt{3}}{2} + 0 \cdot \frac{1}{2} \end{pmatrix} \\ &= \begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \end{aligned}$$

(ii) Let

$$A = \begin{pmatrix} 2 & 5 \\ 1 & -1 \end{pmatrix}$$

The determinant of A is

$$\det(A) = 2(-1) - 5(1) = -7$$

A transformation by a matrix scales area by the factor $|\det(A)|$, so the area scale factor is

$$|-7| = 7$$

Hence the area of triangle T is

$$\frac{84}{7} = 12$$

Now find the area of T in terms of k .

Using the vertex $(1, 2)$, the side vectors are

$$\begin{pmatrix} 5-1 \\ 5-2 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \quad \begin{pmatrix} k-1 \\ -1-2 \end{pmatrix} = \begin{pmatrix} k-1 \\ -3 \end{pmatrix}$$

So

$$\begin{aligned} \text{Area of } T &= \frac{1}{2} \left| \det \begin{pmatrix} 4 & k-1 \\ 3 & -3 \end{pmatrix} \right| \\ &= \frac{1}{2} |4(-3) - 3(k-1)| \\ &= \frac{1}{2} |-12 - 3k + 3| \\ &= \frac{1}{2} |-3k - 9| \\ &= \frac{3}{2} |k + 3| \end{aligned}$$

Since $k > -3$, we have $k + 3 > 0$, so

$$|k + 3| = k + 3$$

Therefore

$$\begin{aligned} \frac{3}{2}(k + 3) &= 12 \\ k + 3 &= 8 \\ k &= 5 \end{aligned}$$

Hence the value of k is 5.

2.

$$\mathbf{P} = \begin{pmatrix} p+1 & 2 \\ 2 & p+1 \end{pmatrix}$$

where p is a positive constant.

The matrix $\mathbf{P}$ represents a linear transformation U .

The parallelogram S has vertices at the points with coordinates $(1, 1)$, $(4, 2)$, $(6, 5)$ and $(3, 4)$.

The area of the image of S under the linear transformation U is 35.

(a) Determine the value of p . [4]

The transformation V consists of a reflection in the line $y = x$ followed by a stretch of scale factor 2 parallel to the x -axis with the y -axis invariant. The transformation V is represented by the matrix $\mathbf{Q}$.

(b) Write down the matrix $\mathbf{Q}$. [2]

Given that U followed by V is the transformation W , which is represented by the matrix $\mathbf{R}$

(c) find the matrix $\mathbf{R}$. [2]

Solution

(a) Take two adjacent sides of the parallelogram from the vertex $(1, 1)$:

$$\begin{pmatrix} 4 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

So the area of S is

$$\left| \det \begin{pmatrix} 3 & 2 \\ 1 & 3 \end{pmatrix} \right| = |3 \cdot 3 - 2 \cdot 1| = |9 - 2| = 7$$

A linear transformation represented by matrix $\mathbf{P}$ multiplies area by the factor $|\det \mathbf{P}|$.

Since the image of S has area 35,

$$35 = |\det \mathbf{P}| \times 7$$

so

$$|\det \mathbf{P}| = 5$$

Now

$$\mathbf{P} = \begin{pmatrix} p+1 & 2 \\ 2 & p+1 \end{pmatrix}$$

so

$$\begin{aligned} \det \mathbf{P} &= (p+1)(p+1) - 2 \cdot 2 \\ &= (p+1)^2 - 4 \\ &= p^2 + 2p + 1 - 4 \\ &= p^2 + 2p - 3 \end{aligned}$$

Hence

$$|p^2 + 2p - 3| = 5$$

So either

$$p^2 + 2p - 3 = 5$$

or

$$p^2 + 2p - 3 = -5$$

From the first:

$$\begin{aligned}p^2 + 2p - 8 &= 0 \\(p + 4)(p - 2) &= 0\end{aligned}$$

so

$$p = -4 \quad \text{or} \quad p = 2$$

From the second:

$$p^2 + 2p + 2 = 0$$

which has no real solutions.

Since p is positive,

$$p = 2$$

(b) Reflection in the line $y = x$ has matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

A stretch of scale factor 2 parallel to the x -axis, with the y -axis invariant, has matrix

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

Since the reflection is followed by the stretch,

$$\mathbf{Q} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$

(c) From part (a), $p = 2$, so

$$\mathbf{P} = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$$

Since U followed by V is represented by $\mathbf{R}$,

$$\mathbf{R} = \mathbf{QP}$$

Therefore

$$\begin{aligned}\mathbf{R} &= \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix} \\&= \begin{pmatrix} 0 \cdot 3 + 2 \cdot 2 & 0 \cdot 2 + 2 \cdot 3 \\ 1 \cdot 3 + 0 \cdot 2 & 1 \cdot 2 + 0 \cdot 3 \end{pmatrix} \\&= \begin{pmatrix} 4 & 6 \\ 3 & 2 \end{pmatrix}\end{aligned}$$

3. (a) Specify fully the transformation $\mathbf{T}$ of the plane associated with the matrix $\mathbf{M}$, where

$$\mathbf{M} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

and $0 < \theta < 2\pi$.

[2]

- (b) (i) Find $\det \mathbf{M}$.

[1]

- (ii) Deduce **two properties** of the transformation $\mathbf{T}$ from the value of $\det \mathbf{M}$.

[2]

- (c) Prove that

$$\mathbf{M}^n = \begin{pmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{pmatrix}$$

where n is a positive integer.

[4]

- (d) Hence specify fully a **single** transformation which is equivalent to n applications of the transformation $\mathbf{T}$.

[1]

Solution

- (a) For a point with position vector $\begin{pmatrix} x \\ y \end{pmatrix}$, the transformation gives

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

This is the standard matrix for a rotation in the plane.

So $\mathbf{T}$ is a rotation through angle θ anticlockwise about the origin.

- (b) (i)

$$\begin{aligned} \det \mathbf{M} &= \begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix} = \cos \theta \cos \theta - (-\sin \theta)(\sin \theta) \\ &= \cos^2 \theta + \sin^2 \theta = 1 \end{aligned}$$

- (ii) Since $|\det \mathbf{M}| = 1$, the area scale factor is 1.

So areas are preserved.

Also, $\det \mathbf{M} > 0$, so orientation is preserved.

Therefore there is no reflection.

The two properties are: area is preserved, and orientation is preserved.

- (c) We prove the result by induction on n .

First, when $n = 1$,

$$\mathbf{M}^1 = \mathbf{M} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} \cos(1\theta) & -\sin(1\theta) \\ \sin(1\theta) & \cos(1\theta) \end{pmatrix}$$

So the statement is true for $n = 1$.

Now assume that for some positive integer k ,

$$\mathbf{M}^k = \begin{pmatrix} \cos(k\theta) & -\sin(k\theta) \\ \sin(k\theta) & \cos(k\theta) \end{pmatrix}$$

Then

$$\mathbf{M}^{k+1} = \mathbf{M}^k \mathbf{M}$$

Substituting the induction hypothesis,

$$\mathbf{M}^{k+1} = \begin{pmatrix} \cos(k\theta) & -\sin(k\theta) \\ \sin(k\theta) & \cos(k\theta) \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Multiplying the matrices,

$$\mathbf{M}^{k+1} = \begin{pmatrix} \cos(k\theta) \cos \theta - \sin(k\theta) \sin \theta & -\cos(k\theta) \sin \theta - \sin(k\theta) \cos \theta \\ \sin(k\theta) \cos \theta + \cos(k\theta) \sin \theta & -\sin(k\theta) \sin \theta + \cos(k\theta) \cos \theta \end{pmatrix}$$

Using the angle addition formulae,

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

we get

$$\mathbf{M}^{k+1} = \begin{pmatrix} \cos((k+1)\theta) & -\sin((k+1)\theta) \\ \sin((k+1)\theta) & \cos((k+1)\theta) \end{pmatrix}$$

So the statement is true for $n = k + 1$.

If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction

$$\mathbf{M}^n = \begin{pmatrix} \cos(n\theta) & -\sin(n\theta) \\ \sin(n\theta) & \cos(n\theta) \end{pmatrix}$$

for all positive integers n .

(d) From part (c), n applications of $\mathbf{T}$ have matrix

$$\mathbf{M}^n = \begin{pmatrix} \cos(n\theta) & -\sin(n\theta) \\ \sin(n\theta) & \cos(n\theta) \end{pmatrix}$$

This is the matrix of a rotation through angle $n\theta$ anticlockwise about the origin.

So the single equivalent transformation is a rotation about the origin through angle $n\theta$ anticlockwise.

4. Consider the matrix

$$A = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

(i) Calculate A^2 . [1]

You are told that the matrix A represents a reflection in a line through the origin.

(ii) Explain why your answer to part (i) is consistent with this information. [1]

(iii) By finding the invariant points of the transformation, determine the equation of the mirror line. [3]

(iv) Describe fully the transformation represented by

$$B = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} \quad [2]$$

(v) A composite transformation is formed by the transformation represented by A followed by the transformation represented by B . Find the single matrix that represents this composite transformation. [2]

(vi) The composite transformation described in part (v) is equivalent to a single reflection. Find the equation of the mirror line of this reflection. [1]

Solution

(i)

$$A^2 = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

Calculating each entry,

$$\begin{aligned} A^2 &= \begin{pmatrix} \left(-\frac{1}{2}\right)\left(-\frac{1}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) & \left(-\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{2}\right) \\ \left(\frac{\sqrt{3}}{2}\right)\left(-\frac{1}{2}\right) + \left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) & \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{4} + \frac{3}{4} & -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} & \frac{3}{4} + \frac{1}{4} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

(ii) A reflection done twice takes every point back to its original position, so its matrix squared should be the identity matrix.

Since in part (i) we found $A^2 = I$, this is consistent with A representing a reflection.

(iii) Points on the mirror line are unchanged by the reflection, so invariant points satisfy

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

Thus

$$\begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

This gives the simultaneous equations

$$\begin{aligned}-\frac{1}{2}x + \frac{\sqrt{3}}{2}y &= x \\ \frac{\sqrt{3}}{2}x + \frac{1}{2}y &= y\end{aligned}$$

Multiply both equations by 2:

$$\begin{aligned}-x + \sqrt{3}y &= 2x \\ \sqrt{3}x + y &= 2y\end{aligned}$$

So

$$\begin{aligned}-3x + \sqrt{3}y &= 0 \\ \sqrt{3}x - y &= 0\end{aligned}$$

From either equation,

$$y = \sqrt{3}x$$

So the mirror line is

$$y = \sqrt{3}x$$

(iv) The standard matrix for a rotation through angle θ about the origin is

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Now

$$B = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

So

$$\cos \theta = -\frac{1}{2}, \quad \sin \theta = \frac{\sqrt{3}}{2}$$

This gives

$$\theta = 120^\circ$$

Therefore B represents a rotation of 120° anticlockwise about the origin.

(v) If A is followed by B , the composite matrix is BA .

$$\begin{aligned}BA &= \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \\ &= \begin{pmatrix} \left(-\frac{1}{2}\right)\left(-\frac{1}{2}\right) + \left(-\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) & \left(-\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(-\frac{\sqrt{3}}{2}\right)\left(\frac{1}{2}\right) \\ \left(\frac{\sqrt{3}}{2}\right)\left(-\frac{1}{2}\right) + \left(-\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) & \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2}\right)\left(\frac{1}{2}\right) \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{4} - \frac{3}{4} & -\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} & \frac{3}{4} - \frac{1}{4} \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}\end{aligned}$$

(vi) Let

$$M = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

The mirror line consists of invariant points, so solve

$$M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

This gives

$$\begin{aligned} -\frac{1}{2}x - \frac{\sqrt{3}}{2}y &= x \\ -\frac{\sqrt{3}}{2}x + \frac{1}{2}y &= y \end{aligned}$$

Multiplying by 2,

$$\begin{aligned} -x - \sqrt{3}y &= 2x \\ -\sqrt{3}x + y &= 2y \end{aligned}$$

So

$$\begin{aligned} -3x - \sqrt{3}y &= 0 \\ -\sqrt{3}x - y &= 0 \end{aligned}$$

Hence

$$y = -\sqrt{3}x$$

So the mirror line is

$$y = -\sqrt{3}x$$

5. The transformation U , represented by the 2×2 matrix P , is a reflection in the x -axis.

- (a) Write down the matrix P . [1]

The transformation V , represented by the 2×2 matrix Q , is a reflection in the line $y = x$.

- (b) Write down the matrix Q . [1]

Given that U followed by V is transformation T , which is represented by the matrix R

- (c) express R in terms of P and Q , [1]
 (d) find the matrix R , [2]
 (e) give a full geometrical description of T as a single transformation. [2]

Solution

- (a) A reflection in the x -axis maps (x, y) to $(x, -y)$, so

$$P = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- (b) A reflection in the line $y = x$ swaps the coordinates, so

$$Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

- (c) If a point is first transformed by U and then by V , then with column vectors

$$\mathbf{x} \mapsto P\mathbf{x} \mapsto Q(P\mathbf{x}) = (QP)\mathbf{x}$$

Hence

$$R = QP$$

- (d) Using part (c),

$$R = QP = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Multiplying,

$$R = \begin{pmatrix} 0 \cdot 1 + 1 \cdot 0 & 0 \cdot 0 + 1 \cdot (-1) \\ 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 0 \cdot (-1) \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

- (e) Apply R to a general point:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

So $(x, y) \mapsto (-y, x)$.

This is a rotation through 90° anticlockwise about the origin.

Therefore T is a rotation of 90° anticlockwise about the origin.

6. The matrix $\mathbf{M}$ is given by

$$\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$$

where $k > 0$ and $k \neq 1$.

- (a) The matrix $\mathbf{M}$ represents a sequence of two geometrical transformations. State the type of each transformation, and make clear the order in which they are applied. [2]

The unit square in the x - y plane is transformed by $\mathbf{M}$ onto rectangle $OABC$.

- (b) Find, in terms of k , the area of rectangle $OABC$ and the matrix which transforms $OABC$ onto the unit square. [3]

- (c) Show that the line through the origin with gradient $\frac{1}{\sqrt{k}}$ is invariant under the transformation in the x - y plane represented by $\mathbf{M}$. [3]

Solution

- (a) Write

$$\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$$

The right-hand matrix is applied first. Since

$$\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ ky \end{pmatrix}$$

this is a stretch parallel to the y -axis, with scale factor k .

Then

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}$$

which swaps the coordinates, so this is a reflection in the line $y = x$.

Therefore the transformations are: first a stretch in the y -direction by scale factor k , then a reflection in the line $y = x$.

- (b) First find $\mathbf{M}$:

$$\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix} = \begin{pmatrix} 0 & k \\ 1 & 0 \end{pmatrix}$$

The area scale factor of a transformation is $|\det \mathbf{M}|$, so

$$\det \mathbf{M} = \begin{vmatrix} 0 & k \\ 1 & 0 \end{vmatrix} = 0 - k = -k$$

Hence the area scale factor is

$$|\det \mathbf{M}| = k$$

Since the unit square has area 1, the area of rectangle $OABC$ is

$$1 \times k = k$$

To transform $OABC$ back to the unit square, we need $\mathbf{M}^{-1}$. For

$$\mathbf{M} = \begin{pmatrix} 0 & k \\ 1 & 0 \end{pmatrix}$$

$$\mathbf{M}^{-1} = \frac{1}{-k} \begin{pmatrix} 0 & -k \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \frac{1}{k} & 0 \end{pmatrix}$$

Therefore the area of $OABC$ is k , and the required matrix is

$$\begin{pmatrix} 0 & 1 \\ \frac{1}{k} & 0 \end{pmatrix}$$

(c) The line through the origin with gradient $\frac{1}{\sqrt{k}}$ has equation

$$y = \frac{x}{\sqrt{k}}$$

Take any point on this line:

$$\begin{pmatrix} x \\ \frac{x}{\sqrt{k}} \end{pmatrix}$$

Apply $\mathbf{M} = \begin{pmatrix} 0 & k \\ 1 & 0 \end{pmatrix}$:

$$\mathbf{M} \begin{pmatrix} x \\ \frac{x}{\sqrt{k}} \end{pmatrix} = \begin{pmatrix} 0 & k \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ \frac{x}{\sqrt{k}} \end{pmatrix} = \begin{pmatrix} \frac{kx}{\sqrt{k}} \\ x \end{pmatrix} = \begin{pmatrix} \sqrt{k}x \\ x \end{pmatrix}$$

Let the image point be $(X, Y) = (\sqrt{k}x, x)$. Then

$$\frac{X}{\sqrt{k}} = \frac{\sqrt{k}x}{\sqrt{k}} = x = Y$$

so

$$Y = \frac{X}{\sqrt{k}}$$

Therefore the image point also lies on the line $y = \frac{x}{\sqrt{k}}$.

Hence the line through the origin with gradient $\frac{1}{\sqrt{k}}$ is invariant under the transformation represented by $\mathbf{M}$.

7. You are given the matrix $\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$.

(a) Find $\mathbf{A}^5$. [1]

(b) Describe the transformation that $\mathbf{A}$ represents. [2]

The matrix $\mathbf{B}$ represents a reflection in the plane $x = y$.

(c) Write down the matrix $\mathbf{B}$. [1]

The point Q has coordinates $(4, -1, 2)$. The point Q' is the image of Q under the transformation represented by $\mathbf{B}$.

(d) Find the coordinates of Q' . [1]

Solution

(a) First find $\mathbf{A}^2$:

$$\mathbf{A}^2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Now multiply once more:

$$\mathbf{A}^3 = \mathbf{A}^2 \mathbf{A} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{I}$$

So

$$\mathbf{A}^5 = \mathbf{A}^3 \mathbf{A}^2 = \mathbf{I} \mathbf{A}^2 = \mathbf{A}^2$$

Hence

$$\mathbf{A}^5 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

(b) Let a general point be $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$. Then

$$\mathbf{A} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} y \\ z \\ x \end{pmatrix}$$

So $\mathbf{A}$ sends (x, y, z) to (y, z, x) , which is a cyclic permutation of the coordinates.

Also, any point on the line $x = y = z$ is unchanged, so this is a rotation about the line $x = y = z$.

Since $\mathbf{A}^3 = \mathbf{I}$, three applications give a full turn, so the angle of rotation is

$$\frac{360^\circ}{3} = 120^\circ$$

For the sense, $\mathbf{A}$ sends $\mathbf{i} \mapsto \mathbf{k}$, $\mathbf{k} \mapsto \mathbf{j}$ and $\mathbf{j} \mapsto \mathbf{i}$. Viewed from $(1, 1, 1)$ towards the origin, the positive x -, y - and z -axes appear in anticlockwise order, so $x \rightarrow z \rightarrow y$ is clockwise.

Therefore $\mathbf{A}$ represents a rotation of 120° clockwise about the line $x = y = z$, viewed from $(1, 1, 1)$ towards the origin.

- (c) A reflection in the plane $x = y$ swaps the x - and y -coordinates and leaves z unchanged.
Therefore

$$\mathbf{B} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- (d) Apply $\mathbf{B}$ to $Q = (4, -1, 2)$:

$$\mathbf{Q}' = \mathbf{B} \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \\ 2 \end{pmatrix}$$

So the coordinates of Q' are $(-1, 4, 2)$.

8. A right-angled triangle T has vertices $A(-2, 1)$, $B(1, 1)$ and $C(1, 5)$. When T is transformed by the matrix

$$P = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

the image is T' .

- (a) Find the coordinates of the vertices of T' . [2]

- (b) Describe fully the transformation represented by P . [2]

The matrices

$$Q = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}$$

and

$$R = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}$$

represent two transformations. When T is transformed by the matrix QR , the image is T'' .

- (c) Find QR . [2]

- (d) Find the determinant of QR . [2]

- (e) Using your answer to part (d), find the area of T'' . [3]

Solution

- (a) A point (x, y) is represented by the column vector $\begin{pmatrix} x \\ y \end{pmatrix}$.

Applying

$$P = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

gives

$$P \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

So each point (x, y) maps to $(-y, x)$.

For $A(-2, 1)$,

$$A' = (-1, -2)$$

For $B(1, 1)$,

$$B' = (-1, 1)$$

For $C(1, 5)$,

$$C' = (-5, 1)$$

Hence the vertices of T' are $A'(-1, -2)$, $B'(-1, 1)$, $C'(-5, 1)$.

- (b) To identify the transformation, look at what P does to the unit basis vectors.

$$P \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

so the positive x -axis maps to the positive y -axis.

Also,

$$P \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

so the positive y -axis maps to the negative x -axis.

This is a rotation of 90° anticlockwise about the origin.

(c)

$$QR = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}$$

Multiplying row by column:

$$QR = \begin{pmatrix} 2(1) + (-1)(1) & 2(2) + (-1)(0) \\ 0(1) + 3(1) & 0(2) + 3(0) \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 3 & 0 \end{pmatrix}$$

(d) Using

$$QR = \begin{pmatrix} 1 & 4 \\ 3 & 0 \end{pmatrix}$$

its determinant is

$$\det(QR) = 1 \cdot 0 - 4 \cdot 3 = -12$$

(e) The original triangle T is right-angled with side lengths

$$AB = 3, \quad BC = 4$$

so its area is

$$\text{Area of } T = \frac{1}{2} \times 3 \times 4 = 6$$

The magnitude of the determinant gives the area scale factor, so

$$\text{area scale factor} = |\det(QR)| = |-12| = 12$$

Therefore,

$$\text{Area of } T'' = 12 \times 6 = 72$$

9. Given that the matrix $\mathbf{M}$, where

$$\mathbf{M} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

represents a rotation through α° about the y -axis in the positive sense given by the right-hand rule

(a) determine the value of α . [1]

(b) Hence determine the smallest possible positive integer value of k for which $\mathbf{M}^k = \mathbf{I}$. [2]

The 3×3 matrix $\mathbf{N}$ represents a reflection in the plane with equation $x = 0$.

(c) Write down the matrix $\mathbf{N}$. [1]

The point A has coordinates $(1, 2, 3)$.

The point B is the image of the point A under the transformation represented by matrix $\mathbf{M}$ followed by the transformation represented by matrix $\mathbf{N}$.

(d) Show that the coordinates of B are $(2\sqrt{2}, 2, -\sqrt{2})$. [2]

Given that O is the origin

(e) show that, to 3 significant figures, the size of angle AOB is 79.4° . [2]

(f) Hence determine the area of triangle AOB , giving your answer to 3 significant figures. [2]

Solution

(a) Comparing $\mathbf{M}$ with the standard rotation matrix about the y -axis,

$$\begin{pmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{pmatrix}$$

gives

$$\cos \alpha = -\frac{\sqrt{2}}{2}, \quad \sin \alpha = -\frac{\sqrt{2}}{2}$$

So α is in the third quadrant, hence

$$\alpha = 225^\circ$$

(b) $\mathbf{M}$ represents a rotation of 225° , so $\mathbf{M}^k$ represents a rotation of $225k^\circ$.

For $\mathbf{M}^k = \mathbf{I}$, the total rotation must be a multiple of 360° , so

$$225k = 360n$$

for some positive integer n .

Dividing by 45,

$$5k = 8n$$

The smallest positive integer k satisfying this is $k = 8$ (with $n = 5$).

So the smallest possible value of k is 8.

(c) Reflection in the plane $x = 0$ changes (x, y, z) to $(-x, y, z)$, so

$$\mathbf{N} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(d) The point A has position vector

$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

First apply $\mathbf{M}$:

$$\begin{aligned} \mathbf{Ma} &= \begin{pmatrix} -\frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \\ &= \begin{pmatrix} -\frac{\sqrt{2}}{2}(1) - \frac{\sqrt{2}}{2}(3) \\ 2 \\ \frac{\sqrt{2}}{2}(1) - \frac{\sqrt{2}}{2}(3) \end{pmatrix} \\ &= \begin{pmatrix} -\frac{4\sqrt{2}}{2} \\ 2 \\ -\frac{2\sqrt{2}}{2} \end{pmatrix} = \begin{pmatrix} -2\sqrt{2} \\ 2 \\ -\sqrt{2} \end{pmatrix} \end{aligned}$$

Then apply $\mathbf{N}$:

$$\begin{aligned} \mathbf{N}(\mathbf{Ma}) &= \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -2\sqrt{2} \\ 2 \\ -\sqrt{2} \end{pmatrix} \\ &= \begin{pmatrix} 2\sqrt{2} \\ 2 \\ -\sqrt{2} \end{pmatrix} \end{aligned}$$

Hence the coordinates of B are

$$(2\sqrt{2}, 2, -\sqrt{2})$$

(e) The position vectors of A and B are

$$\overrightarrow{OA} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 2\sqrt{2} \\ 2 \\ -\sqrt{2} \end{pmatrix}$$

Using the scalar product,

$$\overrightarrow{OA} \cdot \overrightarrow{OB} = 1(2\sqrt{2}) + 2(2) + 3(-\sqrt{2}) = 4 - \sqrt{2}$$

Also,

$$|\overrightarrow{OA}| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$$

and

$$|\overrightarrow{OB}| = \sqrt{(2\sqrt{2})^2 + 2^2 + (-\sqrt{2})^2} = \sqrt{8 + 4 + 2} = \sqrt{14}$$

So if $\theta = \angle AOB$, then

$$\cos \theta = \frac{\overrightarrow{OA} \cdot \overrightarrow{OB}}{|\overrightarrow{OA}| |\overrightarrow{OB}|} = \frac{4 - \sqrt{2}}{14}$$

Therefore

$$\theta = \cos^{-1} \left(\frac{4 - \sqrt{2}}{14} \right) \approx 79.4^\circ$$

So the size of angle AOB is 79.4° to 3 significant figures.

(f) Using the area formula for a triangle,

$$\text{Area of } \triangle AOB = \frac{1}{2} |\vec{OA}| |\vec{OB}| \sin \angle AOB$$

From part (e), $|\vec{OA}| = |\vec{OB}| = \sqrt{14}$, so

$$\text{Area} = \frac{1}{2} (\sqrt{14})(\sqrt{14}) \sin \theta = 7 \sin \theta$$

Using

$$\cos \theta = \frac{4 - \sqrt{2}}{14}$$

we get

$$\sin \theta = \sqrt{1 - \left(\frac{4 - \sqrt{2}}{14} \right)^2}$$

Hence

$$\begin{aligned} \text{Area} &= 7 \sqrt{1 - \left(\frac{4 - \sqrt{2}}{14} \right)^2} \\ &\approx 6.88 \end{aligned}$$

So the area of triangle AOB is 6.88 square units to 3 significant figures.

10. Bilal has been asked to describe the transformation given by the matrix

$$\begin{pmatrix} \frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \end{pmatrix}$$

He writes his answer as follows:

The transformation is a rotation about the y -axis through an angle of θ , where

$$\cos \theta = \frac{\sqrt{3}}{2} \quad \text{and} \quad \sin \theta = \frac{1}{2}$$

$$\theta = 30^\circ$$

Identify and correct the error in Bilal's work.

[2]

Solution

The standard matrix for a rotation about the y -axis is

$$R_y(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$$

Comparing this with

$$\begin{pmatrix} \frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \end{pmatrix}$$

gives

$$\cos \theta = \frac{\sqrt{3}}{2}, \quad \sin \theta = -\frac{1}{2}$$

So the angle is not 30° , but

$$\theta = -30^\circ$$

equivalently 330° .

Bilal's error is that he ignored the negative sign on the sine term. The transformation is a rotation about the y -axis through 30° clockwise when viewed from the positive y -axis.

11. Three non-singular 3×3 matrices, $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{R}$, are such that

$$\mathbf{R}\mathbf{A} = \mathbf{B}$$

The matrix $\mathbf{R}$ represents a rotation about the x -axis through an angle θ and

$$\mathbf{B} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\sin \theta & \cos \theta \\ 0 & \cos \theta & \sin \theta \end{pmatrix}$$

(a) Show that $\mathbf{A}$ is independent of the value of θ . [3]

(b) Give a full description of the single transformation represented by the matrix $\mathbf{A}$. [1]

Solution

(a) Since

$$\mathbf{R}\mathbf{A} = \mathbf{B}$$

and $\mathbf{R}$ is non-singular, we can multiply on the left by $\mathbf{R}^{-1}$:

$$\mathbf{A} = \mathbf{R}^{-1}\mathbf{B}$$

For a rotation about the x -axis through angle θ ,

$$\mathbf{R} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$$

so

$$\mathbf{R}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix}$$

Now calculate $\mathbf{A}$:

$$\begin{aligned} \mathbf{A} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\sin \theta & \cos \theta \\ 0 & \cos \theta & \sin \theta \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\cos \theta \sin \theta + \sin \theta \cos \theta & \cos^2 \theta + \sin^2 \theta \\ 0 & \sin^2 \theta + \cos^2 \theta & -\sin \theta \cos \theta + \cos \theta \sin \theta \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \end{aligned}$$

This matrix contains no θ , so $\mathbf{A}$ is independent of the value of θ .

Hence

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

(b) The matrix $\mathbf{A}$ maps

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} x \\ z \\ y \end{pmatrix}$$

so it leaves x unchanged and swaps y and z .

Therefore, $\mathbf{A}$ represents a reflection in the plane $y = z$.